# Architecture review: from single-shot Eq. 13 to a recursive active-inference filter

Paradigm-Shift Active Inference — IWAI 2026
companion to `notes/theory_brief.md`

2026-05-09

**Abstract**

The v02 implementation applies Hyland & Albarracin (2025)'s closed-form $\lambda$-conservatism update (Eq. 13) recursively as a sequential filter. This produces unbounded log-posterior accumulation: under steady evidence, $|C_t|$ grows linearly in $t$ until saturation pins every agent into total commitment. A memory-decay patch ($\beta = 0.95$ on $C_{\text{old}}$) bounded the magnitude empirically but smuggled a transition model into the wrong place. This note diagnoses four design shortcuts behind the pathology, reads them through the lenses of three reviewers (Hyland on active inference, Leibo on multi-agent norm dynamics, Tan Zhi-Xuan on Bayesian theory of mind), and lays out a four-piece fix that re-grounds the simulation in proper recursive active inference. Two TikZ flowcharts contrast the current and proposed information loops.

## 1 Where we started

The theory brief (`notes/theory_brief.md`, 2026-05-08) specified a $N = 30$ agent model on a Watts–Strogatz small-world graph. Each agent $i$ carries log-preferences $C_i \in \mathbb{R}^{|\mathcal{O}|}$, edge precisions $\{\gamma_{ij}\}$, delegation $d_i \in [0, 1]$ and cost-of-mind-change $\lambda_i \geq 0$. Two evidence streams (environmental, social) are aggregated into a per-step posterior; the headline empirical target is a $(\lambda, \text{drift rate})$ phase diagram with hysteresis.

The brief offered two candidate $C$-update forms:

**(a) gradient on KL+$\lambda$KL:**
$$C_i^{(t+1)} = \arg\min_C \Big[ \mathrm{KL}\big[q_i^{(t+1)} \,\|\, p(o \mid C)\big] + \lambda_i \, \mathrm{KL}\big[p(o \mid C) \,\|\, p(o \mid C_i^{(t)})\big] \Big], \tag{1}$$

**(b) closed-form Eq. 13:**
$$q^*(s) \propto p(s) \cdot \exp\big[\lambda^{-1} \cdot (-F(s))\big]. \tag{2}$$

(**??**) requires an inner gradient loop per agent per step; (**??**) is closed-form. We chose (b) for v02 of the implementation because it is fast and is the form Hyland publishes.

## 2 Where we got to

The current code path lives in `src/population.py::_observe_jit` (lines 138–180). At each step:

1. **Env channel.** `pymdp.infer_states(obs, empirical_prior=D)` returns a per-agent posterior $p_{\text{env}}$.
2. **Social channel.** A trust-weighted pool over neighbours' previous posteriors gives $p_{\text{social}}$.
3. **Belief update.** The pseudo-recursive form
$$C_t \;=\; \beta\, C_{t-1} \;+\; \lambda^{-1} \big[\, d \log p_{\text{social}} + (1 - d) \, \log p_{\text{env}} \big], \tag{3}$$
with $\beta = 0.95$ added in the most recent slice to bound $|C_t|$.

4. **Action.** $a_i \sim \mathrm{softmax}(C_t)$.
5. **Trust update.** $\gamma_{ij} \mathrel{*}= \exp(-\eta \cdot \mathrm{NLL}_j)$.

The pymdp generative model is intentionally trivial: $A = I_{n_o}$, $B = I_{n_o}$ (single null control), $D$ uniform.

**What works.** Heterogeneous-$\lambda$ slow-drift rollouts produce a visible cascade signature: low-$\lambda$ agents sign-flip first, drag neighbours along on a delay (`notebooks/02_phase_space.ipynb`, §4).

**What doesn't.**

1. Without $\beta$, $|C_t|$ runs to $\pm 10^4$ within $\sim$300 steps under consistent evidence. Posterior is then $\sim$one-hot regardless of subsequent observations: agents cannot recover from the wrong belief.
2. With $\beta = 0.95$, the equilibrium $|C_\infty| = e/(\lambda(1-\beta))$ saturates in $\sim$20 steps regardless of $\lambda$. The phase grid is $\lambda$-flat at the resolution we measured.
3. The $\lambda$-cascade in (§4 of the notebook) is real but lives in a narrow window before saturation.

## 3 Diagnosis: four shortcuts

Figure ?? traces the information loops in the current implementation. Two pieces are flagged on the diagram itself; two more are algebraic, given below the figure.

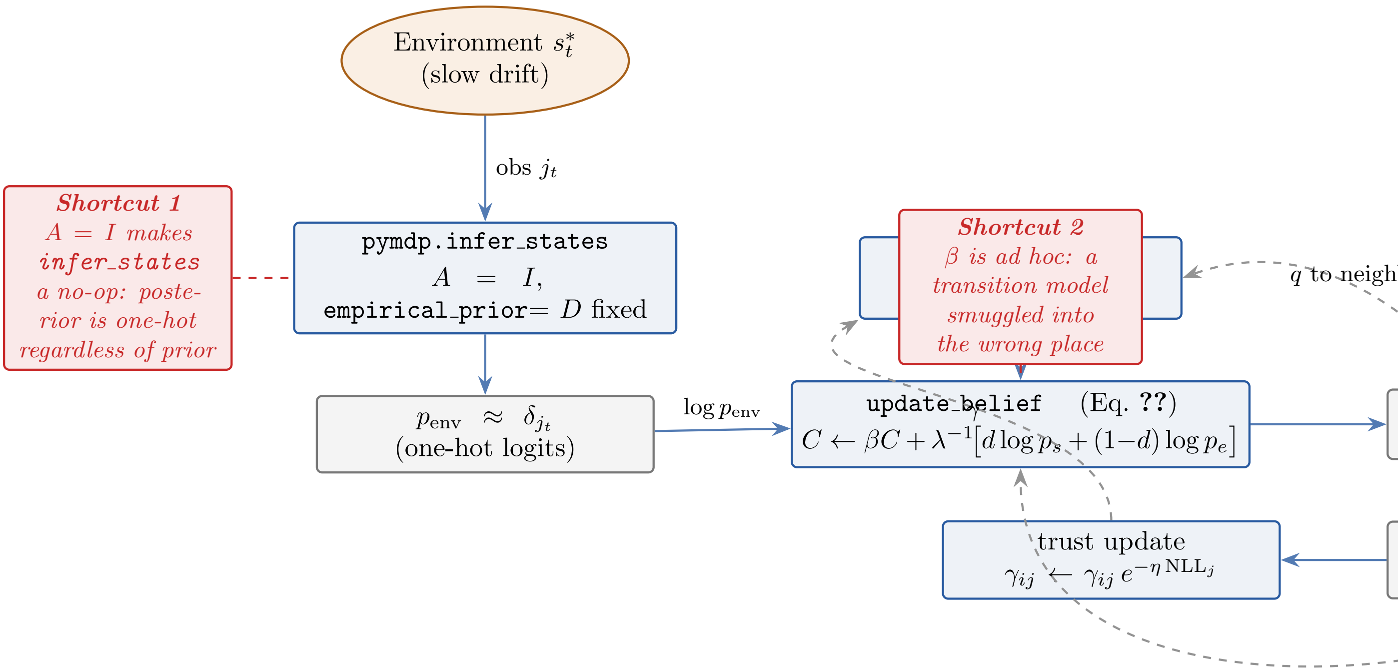


Figure 1: Information flow in the v02 implementation. Solid arrows are within-step computation; dashed arrows are state that feeds back into the next step. Two of the four diagnosed shortcuts are algebraic and shown inline below the figure.

**Shortcut 1: $A = I$ makes `infer_states` a no-op.** With identity observation likelihood, posterior $\propto A[\text{obs}, :] \cdot \text{prior} = e_{\text{obs}} \cdot \text{prior}$, which after normalisation is $\delta_{\text{obs}}$ regardless of the prior. So whatever we feed as `empirical_prior` is overwhelmed by the observation. The pymdp machinery is paying compilation cost for what is mathematically a dictionary lookup.

**Shortcut 2: the $\beta$ patch.** (**??**) with $\beta < 1$ is a linear filter on $C$ with geometric memory. It bounds $|C_t|$ at $e/(\lambda(1-\beta))$ under steady evidence $e$, but the time constant $1/(1-\beta)$ is independent of $\lambda$. $\lambda$ now controls the magnitude of equilibrium confidence, not the speed of belief change. That is not what $\lambda$ is supposed to do in (**??**).

**Shortcut 3: (??) is single-shot.** Hyland & Albarracin's Eq. 13 is a variational solution at one inference step, with $p(s)$ the generative model's prior — fixed. Our recursion $p_t = q_{t-1}$ (i.e. yesterday's posterior is today's prior) is not in the paper. With no transition model in $B$, log-posterior magnitude grows linearly in $t$ forever. This is the actual root cause of the runaway; the $\beta$ patch hides it without addressing it.

**Shortcut 4: the social channel is a flat opinion pool.** $p_{\text{social}} = \sum_j \gamma_{ij}\, q_j$ has no structure: agents do not represent why a neighbour holds their belief. The mixing of $\log p_{\text{env}}$ and $\log p_{\text{social}}$ via $d$ is also an extension to (**??**) that needs justification or it becomes a free parameter that does whatever we want.

## 4 Three perspectives

The four shortcuts are not equally severe under different lenses.

**Hyland (active inference orthodoxy).** Shortcuts 1 and 3 are the load-bearing problems. `pymdp.Agent` already exposes $A$ and $B$; softening both would let pymdp's variational machinery do real recursive filtering, and the $\beta$ patch becomes unnecessary. Hyland would also question the $d$-weighted log-mixing in shortcut 4: there is a coherent two-likelihood form (Section **??**) that respects the variational geometry.

**Leibo (multi-agent / norm dynamics).** The pathology Leibo would flag first is independent of the four shortcuts: *agents observe each other's posteriors, not their actions* (`population.py:158`, where `pop.q_post` flows directly into `neighbour_pool`). In real norm learning, behaviour is observed, not belief: actions are downstream of belief $\times$ utility $\times$ noise, and that noise is what regularises social inference. Removing the action channel removes the regularisation. Compounding this, the trust update reinforces homophily without an exploration counter-pressure: agreement on wrong answers keeps $\gamma$ high; the model has no parameter regime that produces a paradigm shift instead of lock-in, which a model of paradigm shifts ought to.

**Tan Zhi-Xuan (Bayesian ToM).** Each agent should maintain a posterior over each neighbour's belief and reliability, not just a scalar trust. The current $\gamma_{ij}$ is a heuristic decay on outcome agreement; ZX's frame would have $\gamma_{ij}$ derived from predictive-accuracy Bayes, with separate uncertainty about the neighbour's $\lambda$ (deliberate vs. impulsive). This is a substantial structural addition and is out of scope for the current slice.

## 5 Proposed fix

The agreed direction (**Replace $\beta$ with B-matrix drift**) turns out to require four coordinated changes once we look at the actual A/B construction. Each piece is paper-defensible on its own, and together they restore proper recursive active inference.

| Piece | Current | Proposed |
|---|---|---|
| $A$ | $I_{n_o}$ | $(1-\nu)\, I + \nu\, (\mathbf{1}\mathbf{1}^\top - I)/(n_o{-}1)$ |
| $B$ | $I_{n_o}$ | $(1-\varepsilon)\, I + \varepsilon\, \mathbf{1}\mathbf{1}^\top/n_o$ |
| `empirical_prior` | $D$ (fixed uniform) | $q_{\text{post},\, t-1}$ (recursive) |
| update rule | $C_t = \beta\, C_{t-1} + \lambda^{-1}\log\text{evidence}$ | $q_t \propto (B\, q_{t-1}) \cdot p_{\text{env}}^{(1-d)/\lambda} \cdot p_{\text{social}}^{d/\lambda}$ |

Table 1: Four-piece architectural change. New parameters: observation noise $\nu$ (default 0.05), state-drift $\varepsilon$ (default 0.02). The $\beta$ memory parameter is removed.

**Why this works.** Under steady evidence $e$, the proposed update has $q_t \propto B\, q_{t-1} \cdot p_{\text{env}}^{1/\lambda}$. The factor $B\, q_{t-1}$ is bounded below by $\varepsilon/n_o$ in every component, so $\log q_t$ is bounded above. $\lambda$ now controls the *strength* of the likelihood relative to the prior at every step — exactly the role Eq. **??** gives it — without needing memory decay to keep things finite. The temporal coupling lives in $B$, where active-inference theory expects it.

**Equilibrium analysis (sketch).** Fix $d = 0$ (env channel only), steady observation $j$, scalar $\lambda$. The fixed-point posterior $q^*$ satisfies

$$q^*(s) \;\propto\; \big[(1-\varepsilon)q^*(s) + \varepsilon/n_o\big] \cdot p(o = j \mid s)^{1/\lambda},$$

which has a unique attractor with $q^*(j) < 1$ when $\varepsilon > 0$. As $\varepsilon \to 0$ recovery is impossible (the current pathology); as $\varepsilon \to 1$ no learning occurs (uninformative prior every step). $\lambda$ shifts the location of the attractor smoothly: low $\lambda \Rightarrow$ sharper $q^*$, high $\lambda \Rightarrow$ flatter $q^*$. That is what the brief's Predicted Phenomena table assumes.

**The social-channel choice.** The proposed update treats $p_{\text{env}}$ and $p_{\text{social}}$ as two independent likelihoods, each tempered by its own weight on $1/\lambda$. This is mathematically clean ($q_t$ is still proportional to a product of distributions) and reduces to (**??**) when $d = 0$. It changes one thing relative to the current mix: under low $\lambda$, the social channel gets sharpened the same way the env channel does, so cascade signatures should be cleaner, not noisier. We adopt this form.

Figure **??** shows the resulting information flow.

## 6 Mapping back to the original plan

**Verification criteria.** The notebook-level verification criteria in the prior implementation plan (`notebooks/02_phase_space.ipynb` §0–§5) are unchanged in shape. They will be re-checked against the proposed dynamics:

| Criterion | Holds under v02? | Should hold under fix? |
|---|---|---|
| §0 `type(pop.agent).__name__ == "Agent"` | ✓ | ✓ |
| §1 mean-payoff varies smoothly | partially (saturates) | ✓(bounded $\lvert C\rvert$) |
| §2 monotone $(\lambda, \text{drift})$ degradation | *fails* ($\lambda$-flat) | ✓expected: $\lambda$ now controls likelihood weight |
| §3 4 distributions visibly distinct | ✓(in narrow window) | ✓cleaner |
| §4 cascade signature in C-trajectories | ✓ | ✓cleaner |
| §5 dissent front spreading visibly | ✓(after viz fix) | ✓ |

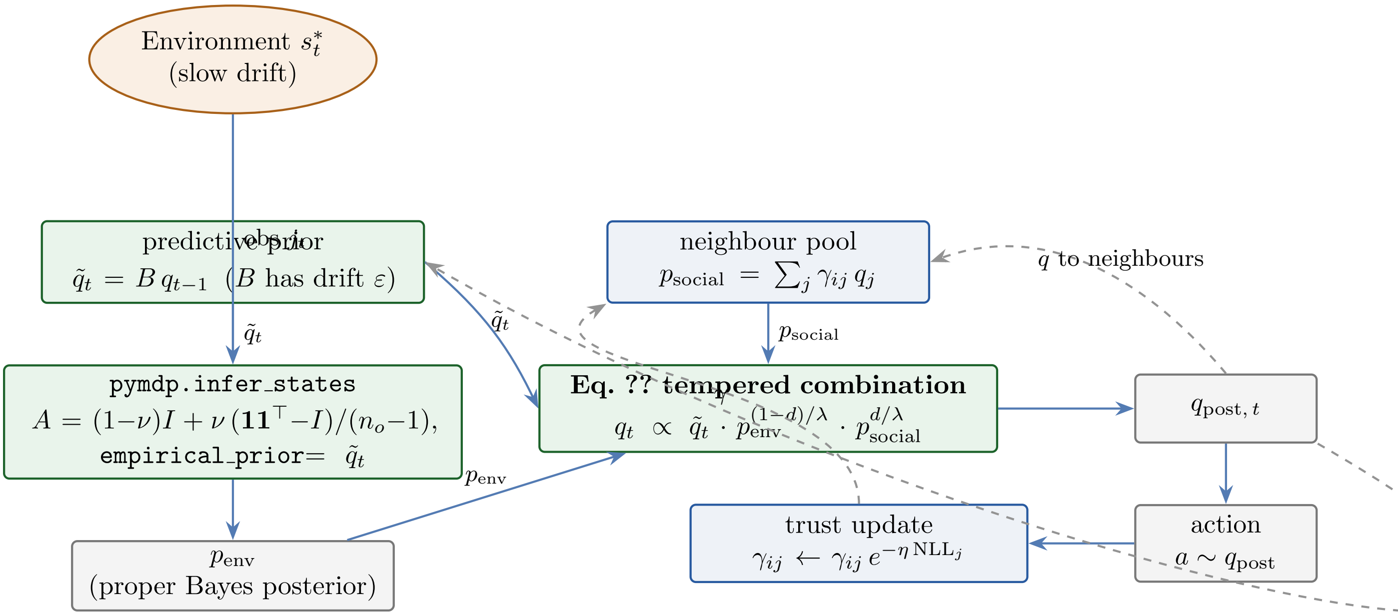


Figure 2: Information flow under the proposed fix. Green boxes are the changed components; the temporal loop now passes through the transition matrix $B$, where active-inference theory expects it. The social and trust loops are unchanged.

**Experimental plan.** The brief's E1–E4 sequence (`notes/theory_brief.md`, §Experimental plan) is unaffected at the *plan* level. The deadlines (E1: 2026-05-14, E2: 2026-05-15–17, E3: 2026-05-18–21, E4: 2026-05-24–27) require the new architecture be in place within ~5 days. The four-piece change is small (one config field, one agent.py function, two lines in population.py setup, two lines in `_observe_jit`); estimated implementation time is one focused session.

**What's still out of scope.** The Leibo and Tan Zhi-Xuan critiques (action-channel observation, theory-of-mind over neighbours) are not addressed by this fix. They are deferred to a follow-up slice if results support the paper's claim. The dissociation experiment vs. rule-induction (E4) does *not* require those critiques to be addressed — it requires the asymmetric $\lambda$-cost mechanism to do meaningful work, which the proposed fix is what enables.

## 7 Decision summary

1. Adopt the four-piece change in Table **??**.
2. Remove `memory_persistence` from `ModelConfig` (it was the $\beta$ patch; no longer needed).
3. Add `obs_noise` ($\nu$, default 0.05) and `state_drift` ($\varepsilon$, default 0.02) to `ModelConfig`.
4. Replace `src/agent.py::update_belief` with the tempered product-of-likelihoods form.
5. Re-execute `notebooks/02_phase_space.ipynb` and re-check verification criteria §1–§5.
6. If §2 still shows $\lambda$-flat behaviour, sweep $\lambda$ over a wider range (e.g. $\{0.1, 0.3, 1.0, 3.0, 10.0\}$ instead of $\{0.1, 0.5, 1.5, 5.0\}$); the change in dynamics may be located outside the previous window.